

Section : Common

Q.1 The area of the region bounded by the parabola $y^2 = 8x$ and its latus rectum in the first quadrant, is

1. $\frac{8}{3}$ sq. units
2. $\frac{32}{3}$ sq. units
3. $\frac{16\sqrt{2}}{3}$ sq. units
4. $\frac{16}{3}$ sq. units

Options 1. 1

2. 2
3. 3
4. 4

Q.2 The function, $f(x) = x - \frac{1}{x}$ is

1. increasing for all $x \in (-\infty, 0) \cup (0, \infty)$
2. decreasing for all $x \in (-\infty, 0) \cup (0, \infty)$
3. increasing for all $x \in (-\infty, \infty)$
4. neither increasing nor decreasing for $x \in (-\infty, \infty)$

Options 1. 1

2. 2
3. 3
4. 4

Q.3 The solution of the differential equation $\frac{dy}{dx} = \sqrt{\frac{y}{x}}$ is

1. $\sqrt{x} + \sqrt{y} = C$: C is an arbitrary constant
2. $\sqrt{x} - \sqrt{y} = C$: C is an arbitrary constant
3. $\sqrt{\frac{y}{x}} + C = 0$: C is arbitrary constant
4. $\sqrt{\frac{x}{y}} + C = 0$: C is arbitrary constant

Options 1. 1

2. 2
3. 3
4. 4

Q.4 If $A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$, $B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$ and $(A + B)^2 = A^2 + B^2$ then

1. $a = 1, b = 4$
2. $a = -1, b = 4$
3. $a = 2, b = -3$
4. $a = 1, b = -4$

Options 1. 1

2. 2
3. 3
4. 4

Q.5 $\int_{\sqrt{\log_e 2}}^{\sqrt{\log_e 4}} x e^{x^2} dx$ is equal to

1. $\frac{1}{2}$
2. 1
3. 2
4. 4

Options 1. 1

2. 2

3. 3

4. 4

Q.6 Given a matrix A of order 3×3 . If $|A| = 3$ then the value of $|A(\text{adj } A)|$ is:

1. 3

2. 27

3. 9

4. 81

Options 1. 1

2. 2

3. 3

4. 4

Q.7 The value of $\begin{vmatrix} x^2 - x + 1 & x - 1 \\ x + 1 & x + 1 \end{vmatrix}$ is equal to

1. $x^3 + x^2 - 2$

2. $x^3 + x^2 + 2$

3. $x^3 - x^2 + 2$

4. $x^3 - x^2 - 2$

Options 1. 1

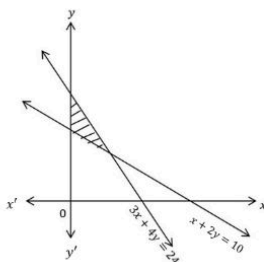
2. 2

3. 3

4. 4

Q.8

For a linear programming problem, the feasible region is shown in the figure by shaded portion, then linear constraints are



1. $3x + 4y \leq 24, x + 2y \leq 10, x \geq 0, y \geq 0$

2. $3x + 4y \geq 24, x + 2y \geq 10, x \geq 0, y \geq 0$

3. $3x + 4y \geq 24, x + 2y \leq 10, x \geq 0, y \geq 0$

4. $3x + 4y \leq 24, x + 2y \geq 10, x \geq 0, y \geq 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.9 If $y = 3e^{2x} + 2e^{3x}$, then $\frac{d^2y}{dx^2} + 6y$ is equal to

1. $5 \frac{dy}{dx}$

2. $6 \frac{dy}{dx}$

3. $-5 \frac{dy}{dx}$

4. y^2

Options 1. 1

2. 2

3. 3

4. 4

Q.10 For $x > 0$, the minimum value of $\frac{x}{\log_e x}$ is

1. $\frac{1}{e}$
2. $2e$
3. e
4. $\frac{2}{e}$

Options 1. 1

2. 2
3. 3
4. 4

Q.11 $\int \frac{1}{x(x^5 - 1)} dx$ is equal to

1. $\frac{1}{5} \log_e \left| \frac{x^5 - 1}{x^5} \right| + C : C \text{ is an arbitrary constant}$
2. $\log_e \left| \frac{x^5 - 1}{x^5} \right| + C : C \text{ is an arbitrary constant}$
3. $\frac{1}{5} (\log_e |x| + \log_e |x^5 - 1|) + C : C \text{ is an arbitrary constant}$
4. $\log_e \left| \frac{x}{x^5 - 1} \right| + C : C \text{ is an arbitrary constant}$

Options 1. 1

2. 2
3. 3
4. 4

Q.12 If $\begin{bmatrix} 2x+1 & 5x \\ 0 & y^2+1 \end{bmatrix} = \begin{bmatrix} x+3 & 10 \\ 0 & 26 \end{bmatrix}$, then the possible values of $x + y$ are:

1. 2 and 5
2. 5 and -1
3. 7 and -3
4. 2 and -5

Options 1. 1

2. 2
3. 3
4. 4

Q.13 For a random variable x , probability distribution $P(x)$ is given by $P(x) = \frac{k}{6}(3 - x)$, $x = 0, 1, 2$, then

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \textbf{List-II}

- (A) k is equal to (I) $\frac{1}{2}$
- (B) $P(x = 0)$ (II) 1
- (C) $P(x < 2)$ (III) $\frac{1}{6}$
- (D) $P(1 < x \leq 2)$ (IV) $\frac{5}{6}$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (I), (B) - (IV), (C) - (III), (D) - (II)
3. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
4. (A) - (II), (B) - (IV), (C) - (I), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.14

The order of $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \left[a \frac{d^2y}{dx^2}\right]^{\frac{1}{3}}$ is

1. 1
2. 2
3. 3
4. 4

Options

1. 1
2. 2
3. 3
4. 4

Q.15 For the L.P.P. Maximize $z = 10x + 6y$ subjected to

$$3x + y \leq 12,$$

$$2x + 5y \leq 34,$$

$$x, y \geq 0.$$

Then the feasible region represented by system of inequalities is

1. Unbounded in first quadrant
2. Bounded in first quadrant
3. Unbounded in second quadrant
4. Not possible (Empty)

Options

1. 1
2. 2
3. 3
4. 4

Section : Applied Mathematics

Q.16 If $x = -4$ is a root of

$$\begin{vmatrix} x & 2 & 3 \\ 1 & x & 1 \\ 3 & 2 & x \end{vmatrix} = 0, \text{ then the sum of the other two roots is}$$

1. 4
2. -3
3. 2
4. 5

Options

1. 1
2. 2
3. 3
4. 4

Q.17

The probability of a shooter hitting the target in one shot is $\frac{1}{4}$

The minimum number of shots needed so that the probability of hitting the target at least once is greater than $\frac{7}{16}$ is:

1. 1
2. 2
3. 3
4. 4

Options

1. 1
2. 2
3. 3
4. 4

Q.18

Evaluate the integral $\int (x^4 + x^2 + 1) d(x^2)$, where C is the constant of integration

1. $x^6 + x^4 + x^2 + C$
2. $\frac{x^6}{6} + \frac{x^4}{4} + \frac{x^2}{2} + C$
3. $\frac{x^6}{3} + \frac{x^4}{2} + x^2 + C$
4. $x^6 + x^5 + x^4 + x^3 + x^2 + x + C$

Options 1. 1

2. 2

3. 3

4. 4

Q.19 Probability distribution of random variable X is

X -2 -1 0 1 2

$P(X)$ 2/11 1/11 4/11 3/11 1/11

Then the value of $E(X)$ is

1. 0

2. $\frac{10}{11}$

3. $-\frac{10}{11}$

4. $\frac{11}{10}$

Options 1. 1

2. 2

3. 3

4. 4

Q.20 A random variable X has the following probability distribution:

X 0 1 2 3

$P(X)$ 0.2 0.1 0.3 0.4

The variance of X will be

1. 4.8

2. 1.29

3. 5.1

4. 4.9

Options 1. 1

2. 2

3. 3

4. 4

Q.21

Evaluate $\int_1^{\sqrt{3}} \frac{1}{1+x^2} dx$

1. $\frac{\pi}{3}$
2. $\frac{2\pi}{3}$
3. $\frac{\pi}{6}$
4. $\frac{\pi}{12}$

Options 1. 1

2. 2
3. 3
4. 4

Q.22

Two pipes A and B can fill a tank respectively in 30 min and 45 min.

Both A and B are opened together for some time and then pipe B is turned off.

If the tank is completely filled in 20 min, find after how many minutes pipe B is turned off.

1. 5 min
2. 30 min
3. 15 min
4. 25 min

Options 1. 1

2. 2
3. 3
4. 4

Q.23

A random sample of 100 individuals provides 25 positive responses.

Then the point estimate of the population proportion with positive responses is:

1. 0.25
2. 0.025
3. 0.125
4. 125

Options 1. 1

2. 2
3. 3
4. 4

Q.24 Match List-I with List-II

List-I

(A) The measurable characteristic of a population is called

(B) The measurable characteristics of a sample is called

(C) A smaller group of a population selected to represent a population is called

(D) The assumption made opposite to the null hypothesis is called

List-II

(I) sample

(II) Alternative hypothesis

(III) Parameter

(IV) statistic

Choose the correct answer from the options given below:

1. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
2. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)
3. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)
4. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Options 1. 1

2. 2
3. 3
4. 4

Q.25

The digit in the unit's place of 6^{500} is:

1. 6

- 2. 5
- 3. 3
- 4. 2

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.26

If the cost function of a product is given by $C(x) = \frac{3}{4}x^2 - 5x + 21$, then the marginal cost when $x = 10$ is

- 1. 10
- 2. 21
- 3. 16
- 4. 5

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.27

Let A and B be square matrices of order 3, then $\det [(A - A^T) + (B - B^T)]$ is equal to

- 1. 0
- 2. 1
- 3. -1
- 4. 3

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.28

The effective rate of return equivalent to a nominal rate of 12% per annum compounded quarterly is:
[Given that $(1.03)^4 = 1.1255$]

- 1. 13.55%
- 2. 12.55%
- 3. 11.55%
- 4. 14.25%

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.29

The corner points of the bounded feasible region for an LPP are $(0, 4)$, $(4, 4)$, $(6, 6)$, $(0, 12)$.

If the objective function is $Z = px + qy$, $p > 0$, $q > 0$, then the condition on p and q so that the maximum of Z occurs at $(6, 6)$ and $(0, 12)$ is

- 1. $p = q$
- 2. $p = 2q$
- 3. $q = 2p$
- 4. $q = 3p$

Options 1. 1

- 2. 2
- 3. 3
- 4. 4

Q.30

Math input error

Math input error

Math input error

Math input error

Math input error

Math input error

Options 1. 1

2. 2

3. 3

4. 4

Q.31 Mean and variance of a binomial distribution are 6 and 2 respectively.

The probability of 2 successes will be

1. $\frac{12}{3^5}$

2. $\frac{16}{3^7}$

3. $\frac{12}{3^7}$

4. $\frac{16}{3^5}$

Options 1. 1

2. 2

3. 3

4. 4

Q.32 Math input error

If he can amortize the balance at 9% per annum compounded monthly for 25 years, then his EMI is:

[Given $(1.0075)^{300} = 9.41$]

Math input error

Math input error

Math input error

Math input error

Options 1. 1

2. 2

3. 3

4. 4

Q.33 If $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $\frac{d^2y}{dx^2}$ is equal to

1. $\frac{b^4}{a^2y^3}$

2. $-\frac{b^4}{a^2y^3}$

3. $\frac{b^2}{a^4y^3}$

4. $-\frac{b^2}{a^4y^3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.34 The equation of the tangent line to $y = 2x^2 + 7$, which is parallel to the line $4x - y + 3 = 0$ is

1. $4x - y + 7 = 0$

2. $4x - y + 3 = 0$

3. $4x - y + 5 = 0$

4. $4x - y + 1 = 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.35 The speed of water current is half the speed of a motor boat.
The motor boat travels 15 km downstream in 1 hour.
Then the time taken to return to the starting point is:

1. 2 hours

2. 3 hours

3. 4 hours

4. 4.5 hours

Options 1. 1

2. 2

3. 3

4. 4

Q.36 A company is shut down due to unavailability of electricity due to non payment of electricity Bill because of some unavoidable circumstances. Under which component of time series does this situation fall?

1. Seasonal

2. Cyclic

3. Irregular

4. Secular

Options 1. 1

2. 2

3. 3

4. 4

Q.37 The function $f : R \rightarrow R$ defined as $f(x) = x^2 + 2x$ is

1. decreasing in $(-\infty, -1]$

2. increasing in $(-\infty, -1]$

3. decreasing in $(-\infty, 2]$

4. increasing on R

Options 1. 1

2. 2

3. 3

4. 4

Q.38 The solution set of the inequality $\frac{2x+3}{x-1} < 0$ is:

1. $x \in (-\infty, 1)$

2. $x \in (-\infty, \frac{3}{2})$

3. $x \in (-\frac{3}{2}, 1)$

4. $x \in (-\infty, \frac{2}{3})$

Options 1. 1

2. 2

3. 3

4. 4

Q.39 For the given five values 3.6, 4.3, 4.3, 3.4, 4.4, the three years moving averages are:

1. 12.2, 12.0, 12.1

2. 4.00, 4.03, 4.40

3. 12.0, 12.1, 13.2

4. 4.067, 4.00, 4.03

Options 1. 1

2. 2

3. 3

4. 4

Q.40 Which of the following statement(s) is/are TRUE?

(A) Skew symmetric matrix of even order is always symmetric

(B) Skew symmetric matrix of odd order is non-singular

(C) Skew symmetric matrix of odd order is singular

(D) Skew symmetric matrix is always square matrix

Choose the correct answer from the options given below:

1. (A) and (D) only

2. (C) and (D) only

3. (B) and (D) only

4. (A) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.41 **Math input error**

[Given that $(1.5)^{1/5} = 1.084$]

1. 11%

2. 10%

3. 8.4%

4. 3.3%

Options 1. 1

2. 2

3. 3

4. 4

Q.42

The minimum value of $\begin{vmatrix} 2 & 2 & 2 \\ 2 & 2+x & 2 \\ 2 & 2 & 2+x \end{vmatrix}, x \in R$ is

1. 1

2. -2

3. 0

4. $-\frac{1}{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.43

If $A = \begin{bmatrix} 2 & 3 & 1 \\ 2 & -1 & 0 \end{bmatrix}$ and $B^T = \begin{bmatrix} 4 & 4 \\ 6 & -2 \\ 2 & 0 \end{bmatrix}$ then $4A + B$ is

1. Not possible

2. $4B$

3. $8A$

4. $3B$

Options 1. 1

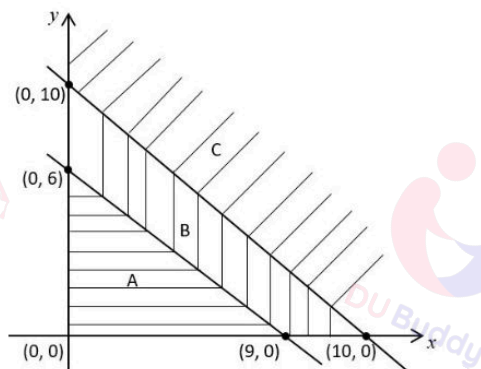
2. 2

3. 3

4. 4

Q.44

In an LPP, the feasible region represented by the set of constraints $2x + 3y \leq 18, x + y \leq 10, x \geq 0, y \geq 0$ is



1. Region A

2. Region B

3. Region C

4. $(\text{Region A}) \cup (\text{Region B})$

Options 1. 1

2. 2

3. 3

4. 4

Q.45 In reference to Inferential Statistics, if $\bar{x}$ is a sample mean of random data $\{x_i\}_{i=1}^n$ and n is the sample size, then the formula

$$\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \text{ represents}$$

1. Mean

2. Variance

3. Standard Deviation

4. t-Statistic

Options 1. 1

2. 2

3. 3

Q.46 The cost of a property appreciates by 10% of the previous month every month.

Math input error

1. end-January 2024
2. end-December 2023
3. end-February 2024
4. end-June 2024

Options 1. 1

2. 2
3. 3
4. 4

Q.47 At what rate of interest will the present value of a perpetuity of

Math input error

1. 10%
2. 12.33%
3. 13.33%
4. 30%

Options 1. 1

2. 2
3. 3
4. 4

Q.48 In a 200 m race, Rohan completes the race in 8 seconds and Vivan completes the race in 10 seconds.
By how much distance Rohan beats Vivan?

1. 30 m
2. 25 m
3. 40 m
4. 35 m

Options 1. 1

2. 2
3. 3
4. 4

Q.49 Math input error

1. 2 : 1
2. 1 : 3
3. 3 : 1
4. 1 : 2

Options 1. 1

2. 2
3. 3
4. 4

Q.50 If a 99% confidence interval states that the population mean is greater than 100 and less than 400,
then the sample mean and margin of error respectively are:

1. 250, 150
2. 150, 250
3. 200, 200
4. 200, 100

Options 1. 1

2. 2

3. 3

4. 4

Section : Core Mathematics

Q.51

Value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \log(\tan x) dx$ is

1. $\frac{\pi}{4}$

2. $\frac{\pi}{12}$

3. 0

4. π

Options 1. 1

2. 2

3. 3

4. 4

Q.52

The general solution of the differential equation $x \left(\frac{dy}{dx} \right) = y + x \tan \left(\frac{y}{x} \right)$ is

1. $\tan \left(\frac{y}{x} \right) = cx$, where c is an arbitrary constant

2. $\sin \left(\frac{y}{x} \right) = cx$, where c is an arbitrary constant

3. $\sin \left(\frac{x}{y} \right) = cx$, where c is an arbitrary constant

4. $\sin \left(\frac{y}{x} \right) = cx^2$, where c is an arbitrary constant

Options 1. 1

2. 2

3. 3

4. 4

Q.53

Two numbers are selected without replacement at random, one at a time from the first six positive integers. Let x denotes the larger of the two numbers.

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \quad \textbf{List-II}

(A) $P(x = 2)$ (I) $\frac{4}{15}$

(B) $P(x = 3)$ (II) $\frac{1}{15}$

(C) $P(x = 4)$ (III) $\frac{2}{15}$

(D) $P(x = 5)$ (IV) $\frac{1}{5}$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

2. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

3. (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

4. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Options 1. 1

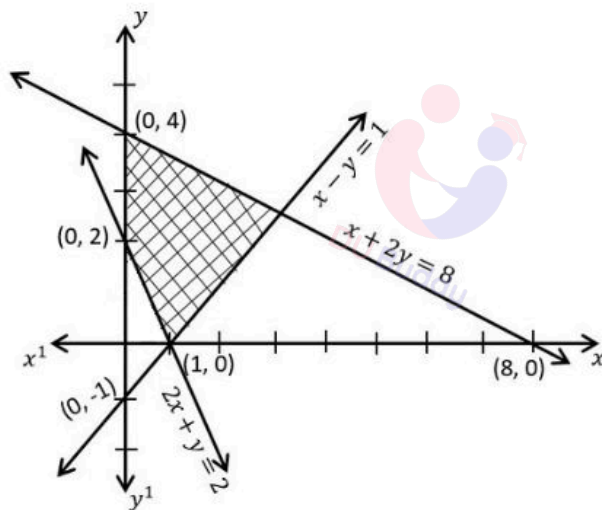
2. 2

3. 3

4. 4

Q.54

The linear inequalities satisfying the shaded feasible region given in the figure are



- (A) $x \geq 0, \quad y \geq 0, \quad 2x + y \geq 2$
 (B) $x \geq 0, \quad y \geq 0, \quad 2x + y \leq 2$
 (C) $x \geq 0, \quad y \geq 0, \quad 2x + y \geq 2, \quad x + 2y \leq 8, \quad x - y \leq 1$
 (D) $x + 2y \geq 8, \quad x - y \geq 1$

Choose the **correct** answer from the options given below:

1. (A) and (C) only
2. (A) and (B) only
3. (B) and (C) only
4. (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.55 The projection vector of the vector $2\hat{i} + 3\hat{j} + \hat{k}$ on $2\hat{i} + \hat{j} - 2\hat{k}$ is

1. $\frac{5}{3}(2\hat{i} + \hat{j} - 2\hat{k})$
2. $\frac{5}{9}(2\hat{i} + \hat{j} - 2\hat{k})$
3. $\frac{5}{14}(2\hat{i} + 3\hat{j} + \hat{k})$
4. $\frac{5}{3}$

Options 1. 1

2. 2
3. 3
4. 4

Q.56 If θ is the angle between two unit vectors $\hat{a}$ and $\hat{b}$ then $|\hat{a} - \hat{b}| =$

1. $\sin \frac{\theta}{2}$
2. $2 \sin \frac{\theta}{2}$
3. $\cos \frac{\theta}{2}$
4. $2 \cos \frac{\theta}{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.57 A spherical ice ball is melting at the rate of $100\pi \text{ cm}^3/\text{min}$. The rate at which its radius is decreasing when its radius is 15 cm, is

1. $\frac{1}{9\pi} \text{ cm/min}$

2. $\frac{1}{9} \text{ cm/min}$

3. $\frac{1}{18} \text{ cm/min}$

4. $\frac{1}{36} \text{ cm/min}$

Options 1. 1

2. 2

3. 3

4. 4

Q.58 The simplified form of $\tan^{-1}\left(\frac{\cos x}{1 + \sin x}\right)$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$ is

1. $\frac{\pi}{4} + \frac{x}{2}$

2. $\frac{\pi}{4} - \frac{x}{2}$

3. $\frac{\pi}{2} + x$

4. $\frac{\pi}{4} + x$

Options 1. 1

2. 2

3. 3

4. 4

Q.59 If $f(x) = \begin{cases} \frac{1-\tan x}{4x-\pi} & x \neq \frac{\pi}{4} \\ k & x = \frac{\pi}{4} \end{cases}$ is continuous at $x = \frac{\pi}{4}$, then the value of k is

1. 1

2. -1

3. $\frac{1}{2}$

4. $-\frac{1}{2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.60 If the system of equations
 $x - 3y + 5z = 3$
 $x - 2y + 4z = 4$
 $2x - 7y + \lambda z = 5$
has infinite number of solutions, then the value of λ is:

1. 2

2. 4

3. 5

4. 11

Options 1. 1

2. 2

3. 3

4. 4

Q.61 The diagonal elements of a skew symmetric matrix are all

- 1.0
- 2.1
- 3.2
- 4.Any real number

Options 1. 1

2. 2
3. 3
4. 4

Q.62

If x, y, z are non-zero numbers, then the inverse of matrix $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ is

1. $\begin{bmatrix} \frac{1}{x} & 0 & 0 \\ 0 & \frac{1}{y} & 0 \\ 0 & 0 & \frac{1}{z} \end{bmatrix}$
2. $\begin{bmatrix} \frac{1}{yz} & 0 & 0 \\ 0 & \frac{1}{xz} & 0 \\ 0 & 0 & \frac{1}{xy} \end{bmatrix}$
3. $\begin{bmatrix} \frac{1}{x^2yz} & 0 & 0 \\ 0 & \frac{1}{xy^2z} & 0 \\ 0 & 0 & \frac{1}{xyz^2} \end{bmatrix}$
4. $\begin{bmatrix} \frac{1}{xyz} & 0 & 0 \\ 0 & \frac{1}{xyz} & 0 \\ 0 & 0 & \frac{1}{xyz} \end{bmatrix}$

Options 1. 1

2. 2
3. 3
4. 4

Q.63 The area of region bounded by the curve $y^2 = 4ax$ and the straight line $x = 2a, a > 0$ in the first quadrant is:

1. $\frac{8a^3}{3}$ sq. units
2. $\frac{8\sqrt{2}a^2}{3}$ sq. units
3. $\frac{32a^3}{3}$ sq. units
4. $\frac{64a^3}{3}$ Sq. units

Options 1. 1

2. 2
3. 3
4. 4

Q.64 If z-coordinate of a point P on the line joining the points A (2, 2, 1) and B (5, 1, -2) is -1, then x-coordinate of point P is

1. 2
2. 4
3. $\frac{4}{3}$
4. -1

Options 1. 1

2. 2
3. 3
4. 4

Q.65 If $y = (x + \sqrt{x^2 + 1})^m$, then $\frac{dy}{dx}$ is

1. $\frac{y}{\sqrt{x^2+1}}$
2. $\frac{m^2y}{\sqrt{x^2+1}}$
3. $\frac{m}{\sqrt{x^2+1}}$
4. $\frac{my}{\sqrt{x^2+1}}$

Options 1. 1

2. 2

3. 3

4. 4

Q.66

The value of $\begin{vmatrix} 265 & 240 & 219 \\ 240 & 225 & 198 \\ 219 & 198 & 181 \end{vmatrix}$ is

1. 5

2. - 4

3. 0

4. 1

Options 1. 1

2. 2

3. 3

4. 4

Q.67

If a line makes angles α, β, γ with the positive directions of the coordinate axes, then the value of $\cos 2\alpha + \cos 2\beta + \cos 2\gamma$ is

1. 1

2. 2

3. - 1

4. - 2

Options 1. 1

2. 2

3. 3

4. 4

Q.68

Nitin has taken the subjects mathematics, physics and chemistry. The probability of him getting grade A in these subjects are respectively 0.2, 0.3 and 0.9 . Getting grades in different subjects are regarded as independent events. The probability of getting A grade by him, either in mathematics or physics, is

1. 0.44

2. 0.5

3. 0.06

4. 0.54

Options 1. 1

2. 2

3. 3

4. 4

Q.69

If A is a singular matrix, then $A(\text{adj } A)$ is equal to

1. a unit matrix

2. a null matrix

3. a symmetric matrix

4. a skew symmetric matrix

Options 1. 1

2. 2

3. 3

4. 4

Q.70

If $A = [a_{ij}]_{3 \times 2}$, where $a_{ij} = i + j$, then

(A) A is a square matrix

(B) $a_{21} + a_{32} = 8$

(C) Number of elements in A is 6

(D) Transpose of A = $\begin{bmatrix} 2 & 3 \\ 3 & 4 \\ 4 & 5 \end{bmatrix}$

Choose the \textbf{correct} answer from the options given below:

1. (A) and (B) only

2. (B) and (C) only

3. (C) and (D) only

4. (A) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.71

The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin |x| + \cos |x|) dx$, is equal to:

1. 0

2. 1

3. 2

4. 4

Options 1. 1

2. 2

3. 3

4. 4

Q.72

The value of derivative of the function $\cot^{-1}\{(\cos 2x)^{1/2}\}$ at $x = \frac{\pi}{6}$ is

1. $\left(\frac{2}{3}\right)^{\frac{1}{2}}$

2. $\left(\frac{1}{3}\right)^{\frac{1}{2}}$

3. $3^{\frac{1}{2}}$

4. $6^{\frac{1}{2}}$

Options 1. 1

2. 2

3. 3

4. 4

Q.73

The corner points of the bounded feasible region determined by the system of linear constraints are $(0, 10), (5, 5), (15, 15), (0, 20)$. Let $z = px + qy$ where $p, q > 0$. Then the condition on p and q so that the maximum value of z occurs at $(15, 15)$ and $(0, 20)$ is

1. $p = q$

2. $p = 2q$

3. $q = 2p$

4. $q = 3p$

Options 1. 1

2. 2

3. 3

4. 4

Q.74

The sum of order and degree of the differential equation $y = x \frac{dy}{dx} + 2\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$ is

1. 1
2. 2
3. 3
4. 4

Options 1. 1

2. 2
3. 3
4. 4

Q.75

If A and B are two events such that $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ and $P(A \cap B) = \frac{1}{4}$, then which of the following statements are true?

(A) A and B are independent events

(B) $P(A|B) = \frac{3}{4}$

(C) $P(A'|B') = \frac{5}{8}$

(D) $P(A'|B) = \frac{1}{4}$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A) and (B) only
4. (B) and (C) only

Options 1. 1

2. 2
3. 3
4. 4

Q.76

Area of the region bounded by the curve $y = \sqrt{x}$ and lines $x + y = 2$, $y = 0$ is

1. $\frac{9}{6}$ square units
2. $\frac{9}{2}$ square units
3. $\frac{7}{6}$ square units
4. $\frac{7}{3}$ square units

Options 1. 1

2. 2
3. 3
4. 4

Q.77

For the function $f(x) = e^x + e^{-x}$

(A) $f'(x) = e^x - e^{-x}$

(B) The critical point is $x = 0$

(C) The minimum value is 1

(D) $x = 0$ is the point of local minimum.

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only
2. (B), (C) and (D) only
3. (C) and (D) only
4. (A) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.78 $\int \frac{\sqrt{16 + (\log x)^2}}{x} dx$ is equal to (where C is an arbitrary constant)

1. $2 \log \left| \log x + \sqrt{16 + (\log x)^2} \right| + \log x \sqrt{16 + (\log x)^2} + C$
2. $16 \log \left| \log x + \sqrt{16 + (\log x)^2} \right| + \frac{\log x}{2} \sqrt{16 + (\log x)^2} + C$
3. $8 \log \left| \log x + \sqrt{16 + (\log x)^2} \right| + \frac{\log x}{2} \sqrt{16 + (\log x)^2} + C$
4. $4 \log \left| \log x + \sqrt{16 + (\log x)^2} \right| + \frac{\log x}{2} \sqrt{16 + (\log x)^2} + C$

Options

1. 1
2. 2
3. 3
4. 4

Q.79 The function $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x|$ ($\mathbb{R}$ is the set of real numbers) is

1. injective but not surjective
2. surjective but not injective
3. both injective and surjective
4. neither injective nor surjective

Options

1. 1
2. 2
3. 3
4. 4

Q.80 The relation R on the set of real numbers defined by $R = \{(a, b) : a \leq b^2\}$ is

- (A) Reflexive
- (B) Not symmetric
- (C) Neither reflexive nor transitive
- (D) Transitive

Choose the \textbf{correct} answer from the options given below:

1. (A) and (D) only
2. (A), (B) and (D) only
3. (B) and (C) only
4. (A) and (C) only

Options

1. 1
2. 2
3. 3
4. 4

Q.81 If the minimum value of a is $-\frac{k}{2}$, such that the function $f(x) = x^2 + ax + 5$ is increasing in $[1, 2]$. Then value of k is

1. -4
2. 2
3. 4
4. -2

Options

1. 1
2. 2
3. 3
4. 4

Q.82 The coordinates of the image of the point $P(5, 4, 2)$ in the line $\vec{r} = (-\hat{i} + 3\hat{j} + \hat{k}) + \lambda(2\hat{i} + 3\hat{j} - \hat{k})$, where λ is a parameter, is

1. $(3, 8, -2)$
2. $(-3, 8, 2)$
3. $(-3, 8, -2)$
4. $(3, 8, 2)$

Options

1. 1

2. 2

3. 3

4. 4

Q.83 If $|\vec{a} - \vec{r}| = |\vec{a}| = |\vec{r}| = 1$, then angle between $\vec{a}$ and $\vec{r}$ is

1. $\frac{\pi}{3}$
2. $\frac{3\pi}{4}$
3. $\frac{\pi}{2}$
4. $\frac{\pi}{6}$

Options 1. 1

2. 2

3. 3

4. 4

Q.84 If A and B are two events such that $P(A) \neq 0$ and $P(B | A) = 1$ then

1. $A \subset B$
2. $B \subset A$
3. $B = \phi$
4. $A = \phi$

Options 1. 1

2. 2

3. 3

4. 4

Q.85 A unit vector perpendicular to the vectors $\hat{i} - \hat{j}$ and $\hat{i} + \hat{j}$ is

1. $\hat{k}$
2. $-\left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$
3. $\frac{\hat{i} - \hat{j}}{\sqrt{2}}$
4. $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

Options 1. 1

2. 2

3. 3

4. 4